

NGTP LITIGATION INTELLIGENCE ENGINE

Comprehensive 13-Step Automated Verification Pipeline & Evidentiary Audit Architecture

Document Classification: Technical Specification & Legal Whitepaper	Target Domain: GST SCNs, DRC-01/07 & Section 16(2)(c) Appeals
Engine Version: v1.0 Production Architecture	Constitutional Framework: Article 141 Supreme Court Precedence

1. Executive Overview & System Philosophy

The NGTP Litigation Intelligence Engine is an autonomous multi-agent system designed to eliminate subjective guesswork, human oversight, and fatal procedural drafting omissions in Indian Goods and Services Tax (GST) litigation. Built specifically for disputes arising under Section 16(2)(c) (supplier tax payment condition) and Section 74 (fraudulent ITC / Non-Genuine Taxpayers), the engine executes a deterministic, 13-step sequential verification pipeline in the background before rendering an actionable verdict (PROCEED vs. HOLD) and synthesizing court-ready IRAC appeal briefs.

Rather than relying on unconstrained LLM text generation, the engine pairs strict statutory rule mathematics with deep dynamic case law calibration under Article 141 of the Constitution of India. The complete 13-step methodology is documented below.

STEP 1: FACT MATRIX & TRANSACTION TIMELINE EXTRACTION	Engine: src/service/fact-matrix-engine.ts
Statutory & Legal Basis: Section 155 CGST Act & Section 106 Indian Evidence Act	Primary Function: Extracts raw factual assertions from Show Cause Notices (DRC-01), Order-in-Original (DRC-07), and taxpayer written submissions. Builds an immutable chronological timeline of the underlying supply.
Methodology & Execution Logic: Cross-references alleged facts against documentary evidence (invoices, bank statements, transport receipts). Flags internal contradictions such as payment dates preceding invoice dates, or supplier GSTIN cancellation predating transaction dates. Categorizes evidentiary strength: Established, Probable, Disputed, or Unsupported.	Status: 100% Background Autonomous
STEP 2: STATUTORY PARAMETERS AUDIT (P1 TO P8 CALIBRATION)	Engine: src/service/statutory-engine.ts
Statutory & Legal Basis: Sections 16(2)(a), 16(2)(b), 16(2)(c), 16(2)(d), 16(2)(aa), 2nd Proviso, 16(4)/(5), and 74(1)	Primary Function: Tests whether the transaction satisfies each of the 8 mandatory statutory conditions governing input tax credit entitlement.
Methodology & Execution Logic: Evaluates: P1 (Tax Invoice under Rule 46); P2 (Actual receipt of goods via E-Way Bills, FASTag, Weighbridge slips); P3 (Tax paid to Government / Suncraft exception); P4 (GSTR-3B return filing); P5 (Prospective GSTR-2B condition w.e.f. 01.01.2022); P6 (Payment within 180 days via bank RTGS); P7 (Section 16(5) retrospective validation up to 30.11.2021); P8 (Extended period of limitation & mens rea test under Section 74).	Status: 100% Background Autonomous
STEP 3: JUDICIAL PRECEDENT RETRIEVAL & REAL-TIME PRECEDENT SYNC	Engine: src/service/precedent-engine.ts & Gemini API Client
Statutory & Legal Basis: Article 141 of the Constitution of India (Law declared by Supreme Court binding on all courts)	Primary Function: Retrieves relevant High Court and Supreme Court authorities governing Section 16(2)(c) recovery from bona fide purchasing recipients.
Methodology & Execution Logic: Continuously queries Google Gemini 1.5 Pro/Flash to identify recently pronounced rulings across all 25 Indian High Courts. Extracts the ratio decidendi, the court's physical evidentiary standard (what documents the judge relied on), and identifies adverse or distinguishable rulings.	Status: 100% Background Autonomous

STEP 4: 6-AXIS PRECEDENT EVIDENTIARY COMPARABILITY & CONFLICT RESOLUTION	Engine: src/service/hierarchy-engine.ts
Statutory & Legal Basis: Judicial discipline & binding precedent doctrine (Union of India v. Raghubir Singh)	Primary Function: Ranks ingested precedents by authority and resolves competing judicial rulings under Article 141.
Methodology & Execution Logic: Computes a Judicial Authority Strength Score (0 to 100) based on bench composition (Supreme Court = 98-100, Division Bench = 86, Single Judge = 74). Applies 6-axis comparability scoring: statutory similarity, factual similarity, evidentiary similarity, procedural similarity, court authority relevance, and distinguishing risk. Automatically resolves conflicts (e.g. Suncraft affirmed by SC overrides adverse rulings like Aastha Enterprises where bank proof was lacking).	Status: 100% Background Autonomous
STEP 5: LOWER AUTHORITY ERROR & JURISDICTIONAL DEFECT AUDIT	Engine: src/service/error-analysis-engine.ts
Statutory & Legal Basis: Section 73/74, Section 75(4) Natural Justice, & CBIC Circulars	Primary Function: Scans the impugned SCN or Order for fatal procedural, evidentiary, and jurisdictional errors committed by the Proper Officer.
Methodology & Execution Logic: Detects recurring administrative flaws: (1) Disallowance of ITC solely on GSTR-2A mismatch in violation of Circular 183; (2) Mechanical recovery from buyer without initiating proceedings against defaulting supplier (violating D.Y. Beathel); (3) Invoking Section 74 extended limitation without establishing fraud or mens rea; (4) Adjudication without mandatory personal hearing under Section 75(4).	Status: 100% Background Autonomous
STEP 6: GROUNDS OF APPEAL OPTIMIZER (IRAC FORMULATION)	Engine: src/service/submission-optimizer.ts
Statutory & Legal Basis: Rule 108 CGST Rules, 2017 & Appellate Tribunal practice	Primary Function: Converts raw factual contentions into structured, court-ready legal grounds of appeal.
Methodology & Execution Logic: Restructures every argument into the formal IRAC legal paradigm: Issue (exact legal error), Rule (statutory section & binding circular), Application (taxpayer's documented facts), and Conclusion (prayer for quashing). Fortifies each ground with Supreme Court SLP order citations and calculates ground strength percentage.	Status: 100% Background Autonomous
STEP 7: ADVERSARIAL RED-TEAM DEFENSE WAR ROOM	Engine: src/service/adversarial-redteam-engine.ts
Statutory & Legal Basis: Revenue Department Standing Counsel adversarial modeling	Primary Function: Simulates the toughest counter-arguments the Revenue will advance during oral hearings before the Appellate Authority or High Court.
Methodology & Execution Logic: Subject the taxpayer's case to aggressive challenges: (1) Condition precedent strictly construed under ALD Automotive; (2) Inability to verify supplier existence; (3) Circular 183 inapplicable due to lack of supplier CA certification; (4) Collusion allegation. Tests whether the defense survives and provides pre-formulated rebuttals.	Status: 100% Background Autonomous
STEP 8: EVIDENCE GAP PRIORITIZATION & ACTION PLAN (P0 / P1 / P2)	Engine: src/service/evidence-gap-engine.ts
Statutory & Legal Basis: Evidentiary threshold management	Primary Function: Identifies missing or corroborative documents needed to elevate case viability before filing.
Methodology & Execution Logic: Classifies missing items into three strict operational tiers: P0 Fatal (Must fix before filing, e.g. bank RTGS proof); P1 Strongly Recommended (Substantially enhances survival probability, e.g. E-Way Bills & FASTag records); P2 Corroborative (Additional strengthening, e.g. CA Certificate under Circular 183).	Status: 100% Background Autonomous
STEP 9: QUANTITATIVE LITIGATION READINESS SCORING (0 TO 100)	Engine: src/service/scoring-engine.ts

Statutory & Legal Basis: Strict evidentiary mathematical weighting	Primary Function: Calculates the objective litigation readiness score from first principles without artificial inflation.
Methodology & Execution Logic: Computes score via weighted matrix: Rule 46 Tax Invoice (15 pts), Bank Payment & 180-Day RTGS Clearance (25 pts), Physical Transit & E-Way Bills (20 pts), Return Filing Reconciliation (15 pts), Retrospective Safe Harbor / Limitation (15 pts), Absence of Fraud / Section 74 Defence (10 pts). Penalizes unverified or missing documents.	Status: 100% Background Autonomous
STEP 10: LITIGATION VIABILITY SCORING & FORUM OUTCOME PROBABILITY	Engine: src/service/scoring-engine.ts
Statutory & Legal Basis: Probabilistic appellate modeling	Primary Function: Estimates the statistical probability of a favorable outcome across Appellate Authority (Section 107), GSTAT (Section 112), and High Court (Article 226).
Methodology & Execution Logic: Synthesizes statutory readiness, precedent authority strength, lower authority error vulnerability, and judicial trend into a viability score (0 to 100) and probability band: HIGH (>75%), MODERATE (50-75%), or LOW (<50%).	Status: 100% Background Autonomous
STEP 11: FORWARD LITIGATION DECISION MATRIX	Engine: src/service/scoring-engine.ts
Statutory & Legal Basis: Risk-adjusted commercial decision theory	Primary Function: Formulates the definitive actionable recommendation for senior tax executives and legal counsels.
Methodology & Execution Logic: Outputs one of four unambiguous decisions: PROCEED (dossier is fully fortified), PROCEED AFTER RECTIFICATION (proceed only after procuring P0/P1 documents), HOLD (untenable in present form; high risk of bank attachment or adverse precedent), or DO NOT PROCEED (fatal defects / fraudulent supply trail).	Status: 100% Background Autonomous
STEP 12: DRAFTING TRAP & DEFECT AUDIT	Engine: src/service/draft-audit-engine.ts
Statutory & Legal Basis: Pleading verification & procedural compliance	Primary Function: Audits taxpayer's written submission, reply, or memo of appeal for catastrophic drafting errors.
Methodology & Execution Logic: Flags fatal drafting traps: (1) Inadvertently conceding that the supplier failed to pay tax; (2) Failing to plead the doctrine of impossibility (lex non cogit ad impossibilia under Arise India); (3) Failing to invoke binding Circular 183; (4) Omitting express prayer for waiver of Section 50 interest and Section 73 penalty.	Status: 100% Background Autonomous
STEP 13: EXECUTIVE VERDICT & APPEAL DOSSIER ASSEMBLY	Engine: src/service/evaluator-agent.ts
Statutory & Legal Basis: Comprehensive litigation dossier compilation	Primary Function: Synthesizes the outputs of all preceding 12 steps into a unified, 1-page executive summary view and exportable court-ready litigation brief.
Methodology & Execution Logic: Generates downloadable legal dossiers formatted for statutory filings (Form GST APL-01) or High Court Writ Petitions, complete with Article 141 case index, IRAC grounds, evidentiary concordance tables, and oral argument notes.	Status: 100% Background Autonomous

2. Summary Architecture & Decision Matrix

Step #	Verification Phase	Core Law / Section	Output Metric
1	Fact Matrix & Timeline	Sec 155 CGST / Evidence Act	Reconciled Timeline & Contradiction Report
2	Statutory Parameters	Sec 16(2)(a)-(d), 16(5)	P1 to P8 Compliance Matrix
3	Precedent Retrieval	Article 141 Constitution	Supreme Court & HC Authorities
4	Comparability & Hierarchy	Bench Strength Doctrine	0-100 Authority Score & Conflict Resolution

5	Lower Authority Errors	Sec 73/74, Sec 75(4)	Jurisdictional Error Dossier
6	Grounds Optimizer	IRAC Formulation	Court-Ready Grounds of Appeal
7	Adversarial Red-Team	Revenue Counsel Simulation	Counter-Party Vulnerability Test
8	Evidence Gaps	Evidentiary Thresholds	P0 / P1 / P2 Actionable Gaps
9	Litigation Readiness	Evidentiary Mathematical Math	0-100 Readiness Score
10	Litigation Viability	Appellate Probability Model	0-100 Viability & Outcome Probability
11	Forward Decision	Risk-Adjusted Decision Matrix	PROCEED / HOLD / RECTIFY Verdict
12	Drafting Audit	Pleading Defect Detection	Fatal Trap Warnings & Corrections
13	Executive Dossier	APL-01 / Writ Brief Assembly	Complete Downloadable Legal Brief